intelligence from it. The hardware used in this level consists of constrained compute devices and gateways. Then comes Level 4, which consists of production planning, logistics, supply chain management and all enterprise applications. All these levels of manufacturing are bound together by data. Data is the single common digital thread that ties the manufacturing process vertically and horizontally, aiding in value generation.

All the data generated thus is stored in the cloud. Operations like data analysis and API management are done at the cloud level. Microservices like measuring, event management and observability also take place in the cloud.

Then comes the user who is at the centre of everything that is being done. Users (whether it is the CEO, a plant manager, or a machine operator) are responsible for the success or failure of the product that a factory is building. These users consume data at different levels.

A smart factory uses a large number of components and each of these components has hundreds of vendors. So a common language, that everyone can understand and work on, is needed. This is called the digital thread, and data-interoperability becomes critical here. It defines the way in which two systems talk to each other. If these systems are unable to communicate with each other, then the entire system fails. This is where open source and the open source community become important in the smart manufacturing space.

The role of a community (common platform)

This common language needs an architecture, and this can most optimally be defined by an open source community. This architecture helps all the components of the smart factory ecosystem to speak and understand the same language. This architecture is also critical to manufacturing better products and processes for the end user.

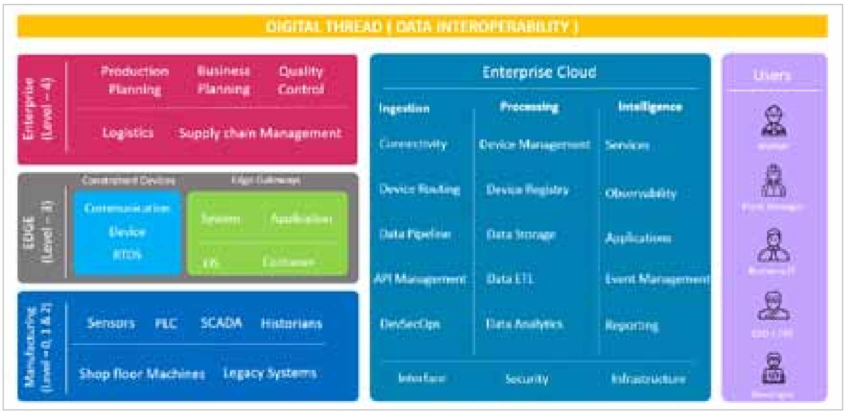

Figure 1: Smart factory ecosystem

Let us take the example of a leading consumer appliance company that offers lifestyle devices that can be customised as per the needs of the end users. So, here, Industry 3.0, which is a tenet of mass manufacturing, is replaced by Industry 4.0. This basically means that the user is at the centre, and defines what he or she wants. All the components are also chosen according to what the user needs.

This company builds an interaction platform to connect end users to its products, and gives them the power to use this platform to define their needs. Then the company uses the same platform to suggest which product will match the requirements of the user. For example, the capacity of an air conditioner can be designed optimally based on the size of the room. The platform gives users the ability to customise these lifestyle goods as per their needs, choice and preference.

So, instead of manufacturing a conventional product which may not be relevant to end users, this company builds products that are defined by them. It builds a platform where all the components of the product can be 3D printed. This is where additive manufacturing and 3D manufacturing are combined into the entire ecosystem. This is the power of a common system, where the user is at the core of the entire ecosystem.

The power of having a common data layer built into the platform enables the company to skip the middlemen, as it is able to directly ship the products from its factory to the end user.

Another interesting case study is about an India based confectionery manufacturer. The chocolates that this confectioner is producing are temperature-sensitive. Refrigeration of these chocolates at the retail point is of utmost importance to the brand for the consumer experience, sales and market capture.

The company is not very sure how the entire ecosystem will play out if it invests in building a common platform. At present, the compressor is being sold to the cooler manufacturer, who is selling the cooler to the confectionery manufacturer. The latter then labels it and passes it on to the retailers. There is no mechanism to find out what the retailers are storing in these coolers and whether they are even placing them inside their shops.

Now, if a consumer sees brand A's products being stored in brand B's cooler, the whole investment in getting a cooler and branding goes to waste. The customer not only gets confused but might end up buying Brand A's products. Moreover, sales can also get affected if the cooler is not functioning properly. A common platform can be of great help in such a scenario.